

desired cohomology is needed. To find out more about this the reader can begin by consulting [Kane 1988] and [Notbohm 1999].

degrees		primes	
$d, 2d, \dots, (n-1)d, nq$ with $q \mid d$		$p \equiv 1 \pmod{d}$ $p \equiv -1 \pmod{d}$	
degrees	primes	degrees	primes
4, 6	$p \equiv 1 \pmod{3}$	60, 60	$p \equiv 1 \pmod{60}$
6, 12	$p \equiv 1 \pmod{3}$	12, 30	$p \equiv 1, 4 \pmod{15}$
4, 12	$p \equiv 1 \pmod{12}$	12, 60	$p \equiv 1, 49 \pmod{60}$
12, 12	$p \equiv 1 \pmod{12}$	12, 20	$p \equiv 1, 9 \pmod{20}$
8, 12	$p \equiv 1 \pmod{4}$	2, 6, 10	$p \equiv 1, 4 \pmod{5}$
8, 24	$p \equiv 1 \pmod{8}$	4, 6, 14	$p \equiv 1, 2, 4 \pmod{7}$
12, 24	$p \equiv 1 \pmod{12}$	6, 9, 12	$p \equiv 1 \pmod{3}$
24, 24	$p \equiv 1 \pmod{24}$	6, 12, 18	$p \equiv 1 \pmod{3}$
6, 8	$p \equiv 1, 3 \pmod{8}$	6, 12, 30	$p \equiv 1, 4 \pmod{15}$
8, 12	$p \equiv 1 \pmod{8}$	4, 8, 12, 20	$p \equiv 1 \pmod{4}$
6, 24	$p \equiv 1, 19 \pmod{24}$	2, 12, 20, 30	$p \equiv 1, 4 \pmod{5}$
12, 24	$p \equiv 1 \pmod{24}$	8, 12, 20, 24	$p \equiv 1 \pmod{4}$
20, 30	$p \equiv 1 \pmod{5}$	12, 18, 24, 30	$p \equiv 1 \pmod{3}$
20, 60	$p \equiv 1 \pmod{20}$	4, 6, 10, 12, 18	$p \equiv 1 \pmod{3}$
30, 60	$p \equiv 1 \pmod{15}$	6, 12, 18, 24, 30, 42	$p \equiv 1 \pmod{3}$

For the prime 2 the realization problem is still not completely solved. The known examples are listed in the short table at the right, where again we give only the irreducible examples, which generate others by taking products. All but the last entry in the table arise from classifying spaces of Lie groups, as described in §4.D. The construction for the last entry is in [Dwyer & Wilkerson 1993].

Lie group	degrees
$O(1)$	1
$SO(n)$	$2, 3, \dots, n$
$SU(n)$	$4, 6, \dots, 2n$
$Sp(n)$	$4, 8, \dots, 4n$
—	$8, 12, 14, 15$

3.H Local Coefficients

Homology and cohomology with local coefficients are fancier versions of ordinary homology and cohomology that can be defined for nonsimply-connected spaces. In various situations these more refined homology and cohomology theories arise naturally and inevitably. For example, the only way to extend Poincaré duality with $\mathbb{Z}$ coefficients to nonorientable manifolds is to use local coefficients. In the overall scheme of algebraic topology, however, the role played by local coefficients is fairly small. Local coefficients bring an extra level of complication that one tries to avoid whenever possible. With this in mind, the goal of this section will not be to give a full exposition but rather just to sketch the main ideas, leaving the technical details for the interested reader to fill in.

The plan for this section is first to give the quick algebraic definition of homology and cohomology with local coefficients, and then to reinterpret this definition more geometrically in a way that looks more like ordinary homology and cohomology. The reinterpretation also allows the familiar properties of homology and cohomology to be extended to the local coefficient case with very little effort.

Local Coefficients via Modules

Let X be a path-connected space having a universal cover $\tilde{X}$ and fundamental group π , so that X is the quotient of $\tilde{X}$ by the action of π by deck transformations $\tilde{x} \mapsto \gamma \cdot \tilde{x}$ for $\gamma \in \pi$ and $\tilde{x} \in \tilde{X}$. The action of π on $\tilde{X}$ induces an action of π on the group $C_n(\tilde{X})$ of singular n -chains in $\tilde{X}$, by sending a singular n -simplex $\sigma: \Delta^n \rightarrow \tilde{X}$ to the composition $\Delta^n \xrightarrow{\sigma} \tilde{X} \xrightarrow{\gamma} \tilde{X}$. The action of π on $C_n(\tilde{X})$ makes $C_n(\tilde{X})$ a module over the group ring $\mathbb{Z}[\pi]$, which consists of the finite formal sums $\sum_i m_i \gamma_i$ with $m_i \in \mathbb{Z}$ and $\gamma_i \in \pi$, with the natural addition $\sum_i m_i \gamma_i + \sum_i n_i \gamma_i = \sum_i (m_i + n_i) \gamma_i$ and multiplication $(\sum_i m_i \gamma_i)(\sum_j n_j \gamma_j) = \sum_{i,j} m_i n_j \gamma_i \gamma_j$. The boundary maps $\partial: C_n(\tilde{X}) \rightarrow C_{n-1}(\tilde{X})$ are $\mathbb{Z}[\pi]$ -module homomorphisms since the action of π on these groups comes from an action on $\tilde{X}$.

If M is an arbitrary module over $\mathbb{Z}[\pi]$, we would like to define $C_n(X; M)$ to be $C_n(\tilde{X}) \otimes_{\mathbb{Z}[\pi]} M$, but for tensor products over a noncommutative ring one has to be a little careful with left and right module structures. In general, if R is a ring, possibly noncommutative, one defines the tensor product $A \otimes_R B$ of a right R -module A and a left R -module B to be the abelian group with generators $a \otimes b$ for $a \in A$ and $b \in B$, subject to distributivity and associativity relations:

- (i) $(a_1 + a_2) \otimes b = a_1 \otimes b + a_2 \otimes b$ and $a \otimes (b_1 + b_2) = a \otimes b_1 + a \otimes b_2$.
- (ii) $ar \otimes b = a \otimes rb$.

In case $R = \mathbb{Z}[\pi]$, a left $\mathbb{Z}[\pi]$ -module A can be regarded as a right $\mathbb{Z}[\pi]$ -module by setting $ay = y^{-1}a$ for $y \in \pi$. So the tensor product of two left $\mathbb{Z}[\pi]$ -modules A and B is defined, and the relation $ay \otimes b = a \otimes yb$ becomes $y^{-1}a \otimes b = a \otimes yb$, or equivalently $a' \otimes b = ya' \otimes yb$ where $a' = y^{-1}a$. Thus tensoring over $\mathbb{Z}[\pi]$ has the effect of factoring out the action of π . To simplify notation we shall write $A \otimes_{\mathbb{Z}[\pi]} B$ as $A \otimes_{\pi} B$, emphasizing the fact that the essential part of a $\mathbb{Z}[\pi]$ -module structure is the action of π .

In particular, $C_n(\tilde{X}) \otimes_{\pi} M$ is defined if M is a left $\mathbb{Z}[\pi]$ -module. These chain groups $C_n(X; M) = C_n(\tilde{X}) \otimes_{\pi} M$ form a chain complex with the boundary maps $\partial \otimes \mathbb{1}$. The homology groups $H_n(X; M)$ of this chain complex are by definition **homology groups with local coefficients**.

For cohomology one can set $C^n(X; M) = \text{Hom}_{\mathbb{Z}[\pi]}(C_n(\tilde{X}), M)$, the $\mathbb{Z}[\pi]$ -module homomorphisms $C_n(\tilde{X}) \rightarrow M$. These groups $C^n(X; M)$ form a cochain complex whose cohomology groups $H^n(X; M)$ are **cohomology groups with local coefficients**.

Example 3H.1. Let us check that when M is a trivial $\mathbb{Z}[\pi]$ -module, with $\gamma m = m$ for all $\gamma \in \pi$ and $m \in M$, then $H_n(X; M)$ is just ordinary homology with coefficients in the abelian group M . For a singular n -simplex $\sigma: \Delta^n \rightarrow X$, the various lifts $\tilde{\sigma}: \Delta^n \rightarrow \tilde{X}$ form an orbit of the action of π on $C_n(\tilde{X})$. In $C_n(\tilde{X}) \otimes_\pi M$ all these lifts are identified via the relation $\tilde{\sigma} \otimes m = \gamma \tilde{\sigma} \otimes \gamma m = \gamma \tilde{\sigma} \otimes m$. Thus we can identify $C_n(\tilde{X}) \otimes_\pi M$ with $C_n(X) \otimes M$, the chain group denoted $C_n(X; M)$ in ordinary homology theory, so $H_n(X; M)$ reduces to ordinary homology with coefficients in M . The analogous statement for cohomology is also true since elements of $\text{Hom}_{\mathbb{Z}[\pi]}(C_n(\tilde{X}), M)$ are functions from singular n -simplices $\tilde{\sigma}: \Delta^n \rightarrow \tilde{X}$ to M taking the same value on all elements of a π -orbit since the action of π on M is trivial, so $\text{Hom}_{\mathbb{Z}[\pi]}(C_n(\tilde{X}), M)$ is identifiable with $\text{Hom}(C_n(X), M)$, ordinary cochains with coefficients in M .

Example 3H.2. Suppose we take $M = \mathbb{Z}[\pi]$, viewed as a module over itself via its ring structure. For a ring R with identity element, $A \otimes_R R$ is naturally isomorphic to A via the correspondence $a \otimes r \mapsto ar$. So we have a natural identification of $C_n(\tilde{X}) \otimes_\pi \mathbb{Z}[\pi]$ with $C_n(\tilde{X})$, and hence an isomorphism $H_n(X; \mathbb{Z}[\pi]) \approx H_n(\tilde{X})$. Generalizing this, let $X' \rightarrow X$ be the cover corresponding to a subgroup $\pi' \subset \pi$. Then the free abelian group $\mathbb{Z}[\pi/\pi']$ with basis the cosets $\gamma\pi'$ is a $\mathbb{Z}[\pi]$ -module and $C_n(\tilde{X}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z}[\pi/\pi'] \approx C_n(X')$, so $H_n(X; \mathbb{Z}[\pi/\pi']) \approx H_n(X')$. More generally, if A is an abelian group then $A[\pi/\pi']$ is a $\mathbb{Z}[\pi]$ -module and $H_n(X; A[\pi/\pi']) \approx H_n(X'; A)$. So homology of covering spaces is a special case of homology with local coefficients. The corresponding assertions for cohomology are not true, however, as we shall see later in the section.

For a $\mathbb{Z}[\pi]$ -module M , let π' be the kernel of the homomorphism $\rho: \pi \rightarrow \text{Aut}(M)$ defining the module structure, given by $\rho(\gamma)(m) = \gamma m$, where $\text{Aut}(M)$ is the group of automorphisms of the abelian group M . If $X' \rightarrow X$ is the cover corresponding to the normal subgroup π' of π , then $C_n(\tilde{X}) \otimes_\pi M \approx C_n(X') \otimes_\pi M \approx C_n(X') \otimes_{\mathbb{Z}[\pi/\pi']} M$. This gives a more efficient description of $H_n(X; M)$.

Example 3H.3. As a special case, suppose that we take $M = \mathbb{Z}$, so $\text{Aut}(\mathbb{Z}) \approx \mathbb{Z}_2 = \{\pm 1\}$. For a nontrivial $\mathbb{Z}[\pi]$ -module structure on M , π' is a subgroup of index 2 and $X' \rightarrow X$ is a 2-sheeted covering space. If τ is the nontrivial deck transformation of X' , let $C_n^+(X') = \{\alpha \in C_n(X') \mid \tau_\#(\alpha) = \alpha\}$ and $C_n^-(X') = \{\alpha \in C_n(X') \mid \tau_\#(\alpha) = -\alpha\}$. It follows easily that $C_n^\pm(X')$ has basis the chains $\sigma \pm \tau\sigma$ for $\sigma: \Delta^n \rightarrow X'$, and we have short exact sequences

$$0 \rightarrow C_n^-(X') \hookrightarrow C_n(X') \xrightarrow{\Sigma} C_n^+(X') \rightarrow 0$$

$$0 \rightarrow C_n^+(X') \hookrightarrow C_n(X') \xrightarrow{\Delta} C_n^-(X') \rightarrow 0$$

where $\Sigma(\alpha) = \alpha + \tau_\#(\alpha)$ and $\Delta(\alpha) = \alpha - \tau_\#(\alpha)$. The homomorphism $C_n(X) \rightarrow C_n^+(X')$ sending a singular simplex in X to the sum of its two lifts to X' is an isomorphism. The quotient map $C_n(X') \rightarrow C_n(X') \otimes_\pi \mathbb{Z}$ has kernel $C_n^+(X')$, so the second short exact sequence gives an isomorphism $C_n^-(X') \approx C_n(X') \otimes_\pi \mathbb{Z}$. These isomorphisms are

isomorphisms of chain complexes and the short exact sequences are short exact sequence of chain complexes, so from the first short exact sequence we get a long exact sequence of homology groups

$$\cdots \rightarrow H_n(X; \tilde{\mathbb{Z}}) \rightarrow H_n(X') \xrightarrow{p_*} H_n(X) \rightarrow H_{n-1}(X; \tilde{\mathbb{Z}}) \rightarrow \cdots$$

where the symbol $\tilde{\mathbb{Z}}$ indicates local coefficients in the module $\mathbb{Z}$ and p_* is induced by the covering projection $p: X' \rightarrow X$.

Let us apply this exact sequence when X is a nonorientable n -manifold M which is closed and connected. We shall use terminology and notation from §3.3. We can view $\mathbb{Z}$ as a $\mathbb{Z}[\pi_1 M]$ -module by letting a loop γ in M act on $\mathbb{Z}$ by multiplication by $+1$ or -1 according to whether γ preserves or reverses local orientations of M . The double cover $X' \rightarrow X$ is then the 2-sheeted cover $\tilde{M} \rightarrow M$ with $\tilde{M}$ orientable. The nonorientability of M implies that $H_n(M) = 0$. Since $H_{n+1}(M) = 0$, the exact sequence above then gives $H_n(M; \tilde{\mathbb{Z}}) \approx H_n(\tilde{M}) \approx \mathbb{Z}$. This can be interpreted as saying that by taking homology with local coefficients we obtain a fundamental class for a nonorientable manifold.

Local Coefficients via Bundles of Groups

Now we wish to reinterpret homology and cohomology with local coefficients in more geometric terms, making it look more like ordinary homology and cohomology.

Let us first define a special kind of covering space with extra algebraic structure. A **bundle of groups** is a map $p: E \rightarrow X$ together with a group structure on each subset $p^{-1}(x)$, such that all these groups $p^{-1}(x)$ are isomorphic to a fixed group G in the following special way: Each point of X has a neighborhood U for which there exists a homeomorphism $h_U: p^{-1}(U) \rightarrow U \times G$ taking each $p^{-1}(x)$ to $\{x\} \times G$ by a group isomorphism. Since G is given the discrete topology, the projection p is a covering space. Borrowing terminology from the theory of fiber bundles, the subsets $p^{-1}(x)$ are called the **fibers** of $p: E \rightarrow X$, and one speaks of E as a bundle of groups with fiber G . It may be worth remarking that if we modify the definition by replacing the word ‘group’ with ‘vector space’ throughout, then we obtain the much more common notion of a vector bundle; see [VBKT].

Trivial examples are provided by products $E = X \times G$. Nontrivial examples we have considered are the covering spaces $M_{\mathbb{Z}} \rightarrow M$ of nonorientable manifolds M defined in §3.3. Here the group G is the homology coefficient group $\mathbb{Z}$, though one could equally well define a bundle of groups $M_G \rightarrow M$ for any abelian coefficient group G .

Homology groups of X with coefficients in a bundle E of abelian groups may be defined as follows. Consider finite sums $\sum_i n_i \sigma_i$ where each $\sigma_i: \Delta^n \rightarrow X$ is a singular n -simplex in X and $n_i: \Delta^n \rightarrow E$ is a lifting of σ_i . The sum of two lifts n_i and m_i of the same σ_i is defined by $(n_i + m_i)(s) = n_i(s) + m_i(s)$, and is also a lift of σ_i . In this way the finite sums $\sum_i n_i \sigma_i$ form an abelian group $C_n(X; E)$, provided we allow the deletion of terms $n_i \sigma_i$ when n_i is the zero-valued lift. A bound-

ary homomorphism $\partial: C_n(X; E) \rightarrow C_{n-1}(X; E)$ is defined by the formula $\partial(\sum_i n_i \sigma_i) = \sum_{i,j} (-1)^j n_i \sigma_i | [v_0, \dots, \hat{v}_j, \dots, v_n]$ where ' n_i ' in the right side of the equation means the restricted lifting $n_i | [v_0, \dots, \hat{v}_j, \dots, v_n]$. The proof that the usual boundary homomorphism ∂ satisfies $\partial^2 = 0$ still works in the present context, so the groups $C_n(X; E)$ form a chain complex. We denote the homology groups of this chain complex by $H_n(X; E)$.

In case E is the product bundle $X \times G$, lifts n_i are simply elements of G , so $H_n(X; E) = H_n(X; G)$, ordinary homology. In the general case, lifts $n_i: \Delta^n \rightarrow E$ are uniquely determined by their value at one point $s \in \Delta^n$, and these values can be specified arbitrarily since Δ^n is simply-connected, so the n_i 's can be thought of as elements of $p^{-1}(\sigma_i(s))$, a group isomorphic to G . However if E is not a product, there is no canonical isomorphism between different fibers $p^{-1}(x)$, so one cannot identify $H_n(X; E)$ with ordinary homology.

An alternative approach would be to take the coefficients n_i to be elements of the fiber group over a specific point of $\sigma_i(\Delta^n)$, say $\sigma_i(v_0)$. However, with such a definition the formula for the boundary operator ∂ becomes more complicated since there is no point of Δ^n that lies in all the faces.

Our task now is to relate the homology groups $H_n(X; E)$ to homology groups with coefficients in a module, as defined earlier. In §1.3 we described how covering spaces of X with a given fiber F can be classified in terms of actions of $\pi_1(X)$ on F , assuming X is path-connected and has the local properties guaranteeing the existence of a universal cover. It is easy to check that covering spaces that are bundles of groups with fiber a group G are equivalent to actions of $\pi_1(X)$ on G by automorphisms of G , that is, homomorphisms from $\pi_1(X)$ to $\text{Aut}(G)$.

For example, for the bundle $M_{\mathbb{Z}} \rightarrow M$ the action of a loop γ on the fiber $\mathbb{Z}$ is multiplication by ± 1 according to whether γ preserves or reverses orientation in M , that is, whether γ lifts to a closed loop in the orientable double cover $\tilde{M} \rightarrow M$ or not. As another example, the action of $\pi_1(X)$ on itself by inner automorphisms corresponds to a bundle of groups $p: E \rightarrow X$ with fibers $p^{-1}(x) = \pi_1(X, x)$. This example is rather similar in spirit to the examples $M_{\mathbb{Z}} \rightarrow M$. In both cases one has a functor associating a group to each point of a space, and all the groups at different points are isomorphic, but not canonically so. Different choices of isomorphisms are obtained by choosing different paths between two points, and loops give rise to an action of π_1 on the fibers.

In the case of bundles of groups $p: E \rightarrow X$ whose fiber G is abelian, an action of $\pi_1(X)$ on G by automorphisms is the same as a $\mathbb{Z}[\pi_1 X]$ -module structure on G .

Proposition 3H.4. *If X is a path-connected space having a universal covering space, then the groups $H_n(X; E)$ are naturally isomorphic to the homology groups $H_n(X; G)$ with local coefficients in the $\mathbb{Z}[\pi]$ -module G associated to E , where $\pi = \pi_1(X)$.*

Proof: As noted earlier, a bundle of groups $E \rightarrow X$ with fiber G is equivalent to an action of π on G . In more explicit terms this means that if $\tilde{X}$ is the universal cover of X , then E is identifiable with the quotient of $\tilde{X} \times G$ by the diagonal action of π , $\gamma(\tilde{x}, g) = (\gamma\tilde{x}, \gamma g)$ where the action in the first coordinate is by deck transformations of $\tilde{X}$. For a chain $\sum_i n_i \sigma_i \in C_n(X; E)$, the coefficient n_i gives a lift of σ_i to E , and n_i in turn has various lifts to $\tilde{X} \times G$. Thus we have natural surjections $C_n(\tilde{X} \times G) \rightarrow C_n(E) \rightarrow C_n(X; E)$ expressing each of these groups as a quotient of the preceding one. More precisely, identifying $C_n(\tilde{X} \times G)$ with $C_n(\tilde{X}) \otimes \mathbb{Z}[G]$ in the obvious way, then $C_n(E)$ is the quotient of $C_n(\tilde{X}) \otimes \mathbb{Z}[G]$ under the identifications $\tilde{\sigma} \otimes g \sim \gamma \cdot \tilde{\sigma} \otimes \gamma \cdot g$. This quotient is the tensor product $C_n(\tilde{X}) \otimes_{\pi} \mathbb{Z}[G]$. To pass to the quotient $C_n(X; E)$ of $C_n(E) = C_n(\tilde{X}) \otimes_{\pi} \mathbb{Z}[G]$ we need to take into account the sum operation in $C_n(X; E)$, addition of lifts $n_i: \Delta^n \rightarrow E$. This means that in sums $\tilde{\sigma} \otimes g_1 + \tilde{\sigma} \otimes g_2 = \tilde{\sigma} \otimes (g_1 + g_2)$, the term $g_1 + g_2$ should be interpreted not in $\mathbb{Z}[G]$ but in the natural quotient G of $\mathbb{Z}[G]$. Hence $C_n(X; E)$ is identified with the quotient $C_n(\tilde{X}) \otimes_{\pi} G$ of $C_n(\tilde{X}) \otimes_{\pi} \mathbb{Z}[G]$. This natural identification commutes with the boundary homomorphisms, so the homology groups are also identified. $\square$

More generally, if X has a number of path-components X_{α} with universal covers $\tilde{X}_{\alpha}$, then $C_n(X; E) = \bigoplus_{\alpha} (C_n(\tilde{X}_{\alpha}) \otimes_{\mathbb{Z}[\pi_1(X_{\alpha})]} G)$, so $H_n(X; E)$ splits accordingly as a direct sum of the local coefficient homology groups for the path-components X_{α} .

We turn now to the question of whether homology with local coefficients satisfies axioms similar to those for ordinary homology. The main novelty is with the behavior of induced homomorphisms. In order for a map $f: X \rightarrow X'$ to induce a map on homology with local coefficients we must have bundles of groups $E \rightarrow X$ and $E' \rightarrow X'$ that are related in some way. The natural assumption to make is that there is a commutative diagram as at the right, such that $\tilde{f}$ restricts to a homomorphism in each fiber. With this hypothesis there is then a chain homomorphism $f_{\#}: C_n(X; E) \rightarrow C_n(X'; E')$ obtained by composing singular simplices with f and their lifts with $\tilde{f}$, hence there is an induced homomorphism $f_*: H_n(X; E) \rightarrow H_n(X'; E')$. The fibers of E and E' need not be isomorphic groups, so change-of-coefficient homomorphisms $H_n(X; G_1) \rightarrow H_n(X; G_2)$ for ordinary homology are a special case. To avoid this extra complication we shall consider only the case that $\tilde{f}$ restricts to an isomorphism on each fiber. With this condition, a commutative diagram as above will be called a **bundle map**.

Here is a method for constructing bundle maps. Starting with a map $f: X \rightarrow X'$ and a bundle of groups $p': E' \rightarrow X'$, let

$$E = \{ (x, e') \in X \times E' \mid f(x) = p'(e') \}.$$

This fits into a commutative diagram as above if we define $p(x, e') = x$ and $\tilde{f}(x, e') = e'$. In particular, the fiber $p^{-1}(x)$ consists of pairs (x, e') with $p'(e') = f(x)$, so $\tilde{f}$ is a bijection of this fiber with the fiber of $E' \rightarrow X'$ over $f(x)$. We use this bijection

to give $p^{-1}(x)$ a group structure. To check that $p:E \rightarrow X$ is a bundle of groups, let $h':(p')^{-1}(U') \rightarrow U' \times G$ be an isomorphism as in the definition of a bundle of groups. Define $h:p^{-1}(U) \rightarrow U \times G$ over $U = f^{-1}(U')$ by $h(x, e') = (x, h'_2(e'))$ where h'_2 is the second coordinate of h' . An inverse for h is $(x, g) \in (x, (h')^{-1}(f(x), g))$, and h is clearly an isomorphism on each fiber. Thus $p:E \rightarrow X$ is a bundle of groups, called the **pullback** of $E' \rightarrow X'$ via f , or the **induced bundle**. The notation $f^*(E')$ is often used for the pullback bundle.

Given any bundle map $E \rightarrow E'$ as in the diagram above, it is routine to check that the map $E \rightarrow f^*(E')$, $e \mapsto (p(e), \tilde{f}(e))$, is an isomorphism of bundles over X , so the pullback construction produces all bundle maps. Thus we see one reason why homology with local coefficients is somewhat complicated: $H_n(X; E)$ is really a functor of two variables, covariant in X and contravariant in E .

Viewing bundles of groups over X as $\mathbb{Z}[\pi_1 X]$ -modules, the pullback construction corresponds to making a $\mathbb{Z}[\pi_1 X']$ -module into a $\mathbb{Z}[\pi_1 X]$ -module by defining $y g = f_*(y) g$ for $f_*:\pi_1(X) \rightarrow \pi_1(X')$. This follows easily from the definitions. In particular, this implies that homotopic maps $f_0, f_1:X \rightarrow X'$ induce isomorphic pullback bundles $f_0^*(E'), f_1^*(E')$. Hence the map $f_*:H_n(X; E) \rightarrow H_n(X'; E')$ induced by a bundle map depends only on the homotopy class of f .

Generalizing the definition of $H_n(X; E)$ to pairs (X, A) is straightforward, starting with the definition of $H_n(X, A; E)$ as the n^{th} homology group of the chain complex of quotients $C_n(X; E)/C_n(A; E)$ where $p:E \rightarrow X$ becomes a bundle of groups over A by restriction to $p^{-1}(A)$. Associated to the pair (X, A) there is then a long exact sequence of homology groups with local coefficients in the bundle E . The excision property is proved just as for ordinary homology, via iterated barycentric subdivision. The final axiom for homology, involving disjoint unions, extends trivially to homology with local coefficients. Simplicial and cellular homology also extend without difficulty to the case of local coefficients, as do the proofs that these forms of homology agree with singular homology for Δ -complexes and CW complexes, respectively. We leave the verifications of all these statements to the energetic reader.

Now we turn to cohomology. One might try defining $H^n(X; E)$ by simply dualizing, taking $\text{Hom}(C_n(X), E)$, but this makes no sense since E is not a group. Instead, the cochain group $C^n(X; E)$ is defined to consist of all functions φ assigning to each singular simplex $\sigma:\Delta^n \rightarrow X$ a lift $\varphi(\sigma):\Delta^n \rightarrow E$. In case E is the product $X \times G$, this amounts to assigning an element of G to each σ , so this definition generalizes ordinary cohomology. Coboundary maps $\delta:C^n(X; E) \rightarrow C^{n+1}(X; E)$ are defined just as with ordinary cohomology, and satisfy $\delta^2 = 0$, so we have cohomology groups $H^n(X; E)$, and in the relative case, $H^n(X, A; E)$, defined via relative cochains $C^n(X, A; E) = \text{Ker}(C^n(X; E) \rightarrow C^n(A; E))$.

For a path-connected space X with universal cover $\tilde{X}$ and fundamental group π , we can identify $H^n(X; E)$ with $H^n(X; G)$, cohomology with local coefficients in the